

Flowing Ink Resonator: An Embodied Soundscape for Culturally Grounded Authoring under Rare Records

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Abstract

Flowing Ink Resonator is an interactive installation in which visitors perform full-body movements in front of a camera to shape a Grand Canal soundscape, while a projection renders visuals, and multichannel audio responds in real time. The system selects and continuously modulates prebuilt soundscape clips and visual “ink seeds” from a corpus derived from literature with provenance links, enabling culturally grounded authoring when direct historical audio records are unavailable or fragmentary. We compare embodied coupling with a button and keyboard baseline that triggers the same events. In a pilot within-subjects (N=6; 12 condition sessions), self report measures and semi structured interviews suggest an exploratory pattern where continuous coupling can shift authoring from discrete triggering toward exploratory modulation, with participants often describing stronger situatedness but also higher learning effort; the baseline clarifies what is achievable via discrete interface control over identical materials. We discuss design implications for embodied cultural soundscapes: culturally legible action schemata, continuity based modulation, and minimal cues for interpretability and uncertainty.

CCS Concepts

• **Human centered computing** → **Interaction paradigms** → **Embodied interaction**;

Keywords

Embodied soundscape installation, Interactive sonification, Heritage sound event corpus, Audiovisual feedback modulation, Presence and sensemaking

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1 Introduction

The Grand Canal carries dense everyday rhythms and sonic memories, but historical canal soundscapes were rarely recorded as audio and are often reduced to generic ambience in exhibitions. We ask how embodied interaction contributes to heritage soundscape experience when historical recordings are incomplete. We present Flowing Ink Resonator, an embodied installation that lets visitors trigger canal sound events through action resembling rowing, with ink like visuals responding to the evolving soundscape. To keep the interaction culturally legible and interpretable, we build a sound event corpus based on literature with traceable provenance and convert sentence level descriptions into callable assets. We evaluate the role of embodied coupling using a control condition where the same events are triggered by buttons, so differences can be attributed to input modality rather than event availability.

Our research questions are: **RQ1**. How does embodied interaction shape presence compared to discrete interface control (button and keyboard input)? **RQ2**. What distinct sensemaking mechanisms does rhythmic bodily effort enable? **RQ3**. What design guidelines arise for embodied heritage soundscapes? As shown in Fig. 1(a), Flowing Ink Resonator links action sensing, sound event triggering and continuous modulation, and ink-wash visuals in a real-time loop. In this project, we contribute: (1) a provenance linked workflow that converts sentence level literary descriptions into a callable sound event and visual gesture asset library for heritage authoring under scarce recordings; (2) an embodied installation that supports continuous scene shaping through rowing like action and real time audiovisual feedback; and (3) an exploratory within subject comparison with a button and keyboard baseline to contextualise how embodied coupling and discrete interface control differently support sensemaking over identical materials. Fig. 1(b) shows the A/B study setup.

2 Related Work

Embodied interaction research shows that meaning can be shaped through sensorimotor coupling, where bodily action participates in perception and interpretation rather than serving only as an input channel. In auditory interaction, sound and soundscape studies further suggest that continuous feedback can support exploratory control, temporal continuity, and spatial anchoring. Mixed reality soundscapes have been used to evoke place based memory and

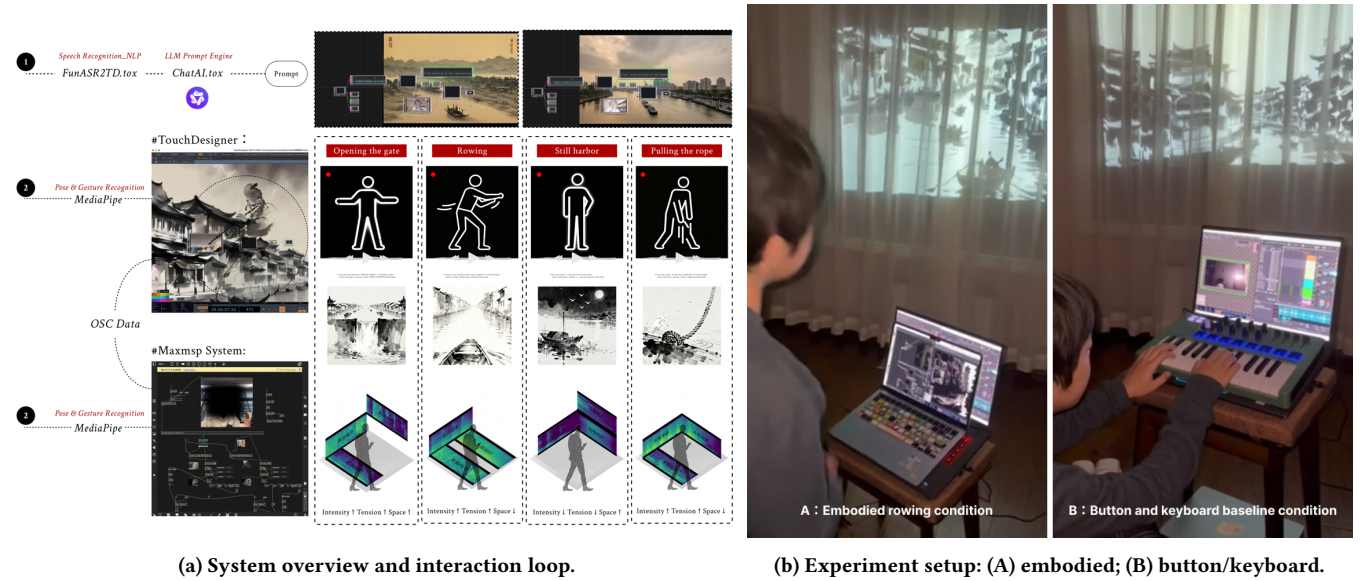


Figure 1: Flowing Ink Resonator overview and A/B study setup.

atmosphere [3], and sound has also been used to examine lived conditions and institutional space [4]. Ecoacoustics and environmental acoustic monitoring extend this view by treating soundscapes as analysable data and exploring them through interactive and visual analytic workflows [1, 6].

In digital heritage, prior work often provides strong accounts of heritage politics, cultural landscapes, and intangible practices, but less often details how textual traces can be operationalised into reusable sound oriented resources for interactive exploration. Grand Canal studies contribute historical and cultural perspectives and examine how cultural practices are communicated [5, 8], while landscape evaluation work adds viewpoints on contemporary heritage value and representation [2]. Recent humanistic AI art and interactive sound research demonstrates how computational pipelines can support culturally situated meaning making and sound interaction design [7, 9].

Direct comparisons between embodied coupling and interface mediated control for heritage soundscape authoring remain limited, especially when immersion, perceived control, and interpretability are treated as linked outcomes. Our work addresses this gap through an exploratory A/B comparison on a literature derived Grand Canal soundscape installation.

3 Methodology

We developed a literature derived soundscape corpus and an interactive installation prototype. We crawled and filtered Grand Canal related text to obtain 4,328 records, from which we extracted 1,120 sentence level sound descriptions as prompts. These units were annotated with scene types, sound source categories, temporal forms, and affect tags. For a set of typical canal scenes, we organized the annotations into scene sound tables to support provenance tracking, linking each invoked resource back to its prompt, semantic unit, and original record. Based on the structured prompt database, we

prebuilt a callable asset library: selected prompts were rendered into reusable sound event clips and ink wash visual seeds. To instantiate assets from text, we generated candidate sound clips from each selected sentence using an audio generation workflow and then curated them through human review to ensure semantic match and cultural plausibility. We trimmed and mixed the accepted clips into short, reusable events, and stored them with metadata for source category, temporal form, and affect tags. Visual “ink seeds” were similarly curated as a small set of conditioning inputs for StreamDiffusion, and the system only composes and modulates these prebuilt assets at runtime. Each asset retains a provenance record linking back to the original sentence, its source record, and annotations. All subsequent interaction operates by selecting, layering, and shaping these prebuilt assets rather than generating new media on the fly.

As shown in Fig. 1a, the installation forms a “virtual boat position” with projection and multichannel audio. TouchDesigner runs the visual engine and renders ink wash imagery using StreamDiffusion conditioned on the currently active visual seeds, while Max/MSP runs the sound engine for playback and mixing of the prebuilt sound event clips. TouchDesigner and Max/MSP exchange control messages via OSC so that sound events and visual responses remain temporally aligned.

Interaction is driven by action sensing and continuous modulation over the same asset set. MediaPipe extracts body keypoints and gesture states from the camera stream; we apply confidence filtering and temporal smoothing to derive stable action states and motion features. These features are sent as OSC to drive two coupled processes: (1) event selection and triggering from the prebuilt sound library, and (2) continuous modulation of playback and visual parameters. The resulting changes update both the evolving soundscape and the ink wash composition in real time, providing immediate audiovisual feedback while participants author a scene.

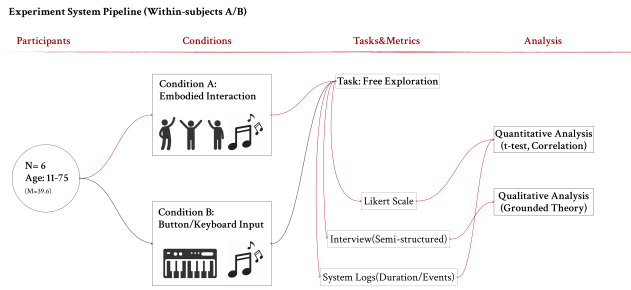


Figure 2: Within-subject A/B study flow: participants, conditions, measures, and analysis.

As shown in Fig. 2, we conducted a within subject A/B study ($N = 6$; 12 sessions). Condition A used full body movement; Condition B used button and keyboard input. Both conditions accessed the same sound events and the same continuous modulation space, differing only in input modality.

Participants. Six volunteers (ages 11–75) completed both conditions, spanning a child, students, working adults (e.g., engineer and clinician), and an older adult (guardian consent for the minor). Given the small and heterogeneous sample, the study is positioned as exploratory and aimed at surfacing interaction patterns rather than establishing generalisable effects

Measures and analysis. After each session, participants completed four 7 point Likert ratings (Control Q2, Presence Q7, Onboarding effort Q12 reverse, Satisfaction Q14) and a semi structured interview; sessions were recorded and transcribed. We used reflexive thematic analysis with constant comparison across conditions: transcripts were open coded into meaning units, codes were grouped into themes, and themes were refined iteratively with representative excerpts. Likert ratings were used only for descriptive triangulation by reporting paired scores and a within participant effect size summary (Cohen’s d_z), without hypothesis testing.

4 Results

We present the interview themes as the main findings and use Likert scale ratings as descriptive triangulation.

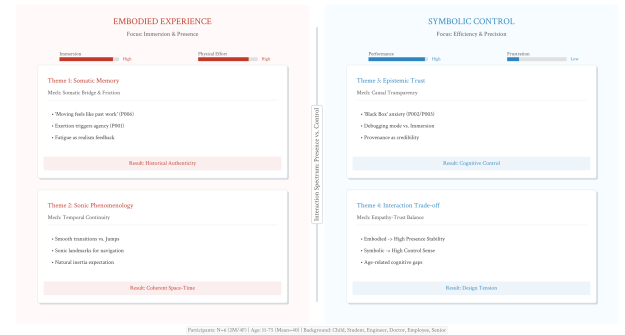


Figure 3: Integrated thematic map comparing embodied interaction (Condition A) and symbolic control (Condition B). Interview codes were grouped into themes with representative evidence, summarising how interaction modality structures heritage soundscape sensemaking.

4.1 Embodied coupling sustains continuity and labour grounded immersion

In Condition A, participants described interaction in action terms and treated sonic change as something they were enacting rather than selecting. Repetition provided a temporal scaffold for attention, supporting continuity interpretation. An older participant related the movement to past work (“moving feels like working as before”), and the child linked exertion to louder outcomes (“when I swing harder, the sound becomes louder”). Likert triangulation aligns with this theme: Presence ratings appear descriptively higher and less variable in the embodied condition.

4.2 Symbolic input increases causal clarity but foregrounds verification

In Condition B, participants emphasised convenience, predictability, and explainability (e.g., “buttons are more convenient”; “otherwise I would start to doubt”). This supported rapid checking by repeating the same input, but also made the interface more salient during authoring. Likert triangulation reflects this emphasis: Control ratings appear descriptively higher in the baseline condition.

4.3 Onboarding is heterogeneous across ages and expectations

Interview excerpts suggest that onboarding depends on expectations and comfort with discrete control. Older adults were sensitive to abrupt change (e.g., “it suddenly becomes loud, it feels strange”), while participants with analytic expectations more readily adopted verification strategies. This matches the paired variability in Onboarding Cost despite similar group means.

5 Discussion

Our study addresses a recurring limitation in cultural soundscape work: historical recordings are sparse, while textual traces are abundant but rarely become operable materials for interactive exploration. By converting literature into a provenance linked corpus and embedding it in an installation, we shift the unit of heritage

sound from “archived audio” to “callable references” that can be invoked, compared, and discussed during authoring. This also clarifies a second gap: comparisons between embodied coupling and button or keyboard control are often confounded by differences in content access or mapping complexity. Here, both conditions access the same sound events and the same continuous modulation space, allowing modality effects to be interpreted more directly.

5.1 From presence to control: what changes when coupling changes

Across findings, embodied coupling primarily supports scene like engagement, while symbolic input primarily supports procedural confidence. With the same sound events available, participants in the embodied condition more often reported presence through continuity oriented exploration: rhythmic effort stabilised attention and gradual change was interpreted as an unfolding canal situation. In the baseline condition, participants more often pursued causal clarity and verification, describing interaction in terms of operating and checking. This pattern accounts for why symbolic control can feel more controllable, while embodied coupling can feel more narratively situated, without reducing the contrast to a simplistic evaluation in terms of better or worse.

5.2 Mechanisms suggested by interview themes

Interview themes suggest two mechanisms that help explain the above contrast. First, rhythmic effort functions as a temporal scaffold that sustains attention during exploration and supports reading change as continuous space time rather than interface feedback. Second, effort becomes an interpretive cue: maintaining motion provides a plausible experiential reason for intensification or approach, reducing the need to articulate explicit rules. By contrast, symbolic input supports a verification scaffold, where repeatable inputs make causal links easier to form and to verbalise, but can also keep attention anchored on the interface.

5.3 Design implications for embodied cultural soundscapes under missing recordings

Three implications follow for systems that reconstruct cultural soundscapes from textual traces. Implication 1, culturally meaningful action schemata: Participants in the embodied condition repeatedly described interaction in role based, action terms, suggesting that mappings aligned with culturally recognisable practices can carry interpretive weight rather than feeling arbitrary. Implication 2, continuity as a plausibility cue: Reports of “unfolding” and rhythm stabilised attention indicate that gradual audiovisual emergence helps participants read change as scene progression instead of interface response. Implication 3, lightweight interpretability cues: Baseline condition talk of “doubting” and “checking” suggests that minimal, non intrusive cues about what is being invoked can reduce rule uncertainty without shifting attention into debugging.

5.4 Broader contribution and significance

Beyond the Grand Canal case, the work outlines a project level workflow for operationalising literature into a provenance linked and reusable soundscape representation, and an interview grounded

account of how coupling modality shapes heritage soundscape sensemaking when content and mapping space are controlled. This positions embodied interaction not only as an input option, but as a narrative and interpretive device for cultural soundscape authoring.

6 Conclusion

Flowing Ink Resonator combines an embodied interaction loop with a literature driven sound event corpus that supports provenance aware assets. With button triggering as a baseline control condition, our exploratory study characterises how embodied rhythm and audiovisual feedback shape continuity oriented sensemaking. The work contributes design guidance for embodied cultural soundscapes when historical audio records are sparse or missing, including culturally grounded action schemata, continuity preserving feedback, and lightweight interpretability cues.

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